

Name: _____

These questions assess your understanding of the material from Chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. Shiv boards a light-rail commuter train each day for work. One day, Shiv decides to use his phone's stopwatch to measure the time it takes for the train to reach its top speed of 92.7 km/h from rest, and finds that it takes 15.6 s. What is the train's average acceleration?

ANSWER: _____

2. Nick is riding a motorcycle that can produce an acceleration of 5.0 m/s^2 while travelling at 100 km/h. The magnitude of the two forces resisting motion, friction and air resistance, total 470 N. The mass of Nick and the motorcycle is 360 kg. What is the magnitude of the force that the motorcycle's rear tire exerts on the ground?

ANSWER: _____

3. Given that the planet Centaurus B orbits the planet Aurora once every 136.6 days and that it is an average distance of 198,000 km from the center of Aurora, calculate the period of an artificial satellite orbiting Aurora at an average altitude of 2,360 km above Aurora's surface. Note that the radius of Aurora is 16,000 km.

ANSWER: _____

4. Ryan, who has a mass of 88 kg, has jumped from a plane and is falling through the air at his terminal velocity, a speed of 55 m/s. The temperature of the air is 11°C . Recall that the density of air ρ (in kg/m^3) depends on temperature T (in $^\circ\text{C}$) according to $\rho = \frac{353}{T+273}$. As Ryan moves over a distance of 8.8 m through the air, calculate his area that faces the air. Assuming a drag coefficient of 0.95.

ANSWER: _____

5. Jennifer wants to use her pressure cooker to cook food in liquid water at a temperature of 129°C . To determine the pressure required inside the pressure cooker for water to boil at this temperature, she can use the Clausius-Clapeyron equation for water: $P(T) = (1 \text{ kPa}) \cdot \exp \left[(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{T} \right) \right]$. In this equation, T is the temperature in Kelvin and P is the pressure in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 37 cm, what is the force that the lid must be able to withstand at the required pressure?

ANSWER: _____

6. A typical small rescue helicopter has four blades, each of which is 3.8-m long and has a mass of 53 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 840 kg. If the helicopter's blades rotate at 251 rpm on the ground, to what height could the helicopter rise if all of its blades' rotational kinetic energy is used to lift it?

ANSWER: _____

7. Daniel wants to place three blocks on a 3.4-kg, 20-m-long wooden board to make it balance atop a triangular wedge of wood. He places a 40-kg purple block a distance of 1.09 m from the left end of the board, a 2.7-kg orange block a distance of 15.6 m from the right end of the board, and places the board on the wedge at a distance of 2.8 m from the left end of the board. Where must he place a 3.9-kg yellow block, measured from the right end of the board, in order to balance the board?

ANSWER: _____

8. A particular hydraulic system contains two fluid-filled cylinders which are connected by a tube. Cylinder 1 has a diameter of 2.1 cm and is capped with piston P_1 . Cylinder 2 has a diameter of 6.2 cm and is capped with piston P_2 . A force F_1 , equal to 120 N, is exerted on P_1 . What is the magnitude of the resulting force exerted by P_2 ?

ANSWER: _____

9. Stephen is an astronaut walking on the surface of the planet Centaurus B. He experiences a small acceleration due to Earth's gravitational field, even at the distance between the center of Earth and Stephen of 0.38 Gm. What is his acceleration due to Earth's gravitational field?

ANSWER: _____

10. Jacob, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.2 cm from the reel's axis of rotation. The reel is given an angular acceleration of 66 rad/s^2 for 2.1 s. How many revolutions does the reel make?
ANSWER: _____
11. Linda attaches a nozzle of diameter 6.0 mm to a garden hose of diameter 1.7 cm. If the hose fills up a 1-L bucket in 11.5 seconds, what is the speed of the water that leaves the nozzle?
ANSWER: _____
12. Leah is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes her and she throws a brick straight down from the bridge to the distant vale below. If the brick falls 39 m in 1.3 s, then how far will it have fallen in 9.0 s?
ANSWER: _____
13. Daniel has to push a 362-kg crate made of aetherium on a floor made of ironbark maple over a distance of 7 m. The crate is initially at rest on the horizontal floor. What is the minimum force that Daniel must exert horizontally on the crate to start moving it? Note that for aetherium on ironbark maple, $\mu_s = 0.27$ and $\mu_k = 0.19$.
ANSWER: _____
14. Eric is at the range firing his M16A2 rifle. When firing a standard 5.56x45mm NATO cartridge, the velocity at which the bullet leaves the muzzle is 3100 ft/s. If Eric were to fire 19.5° above the horizontal, what maximum height would the bullet reach? Report your answer in feet. Ignore air resistance and friction in the barrel. Free-fall acceleration in English units is 32.2 ft/s^2 .
ANSWER: _____
15. A flea jumps into the air by exerting a force straight down on the ground. A breeze blows on the flying flea and exerts a horizontal force of $8 \mu\text{N}$ on the flea. Find the magnitude of the acceleration of the flea if its mass is 0.57 mg.
ANSWER: _____
16. Sophia is driving in her car wondering, how long would it take her, starting from 6.2 m/s and accelerating in a straight line at 1.0 m/s^2 , to cover a distance of 830 m?
ANSWER: _____
17. Adam loads a 78.5-kg package on a roller-belt conveyor system. The package moves at 0.32 m/s. What is its kinetic energy?
ANSWER: _____
18. Laika the moon dog jumps into the air to reach a height of 6.92 m. How long is Laika in the air? Laika is on the Moon and so her free-fall acceleration is 1.625 m/s^2 . Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.
ANSWER: _____
19. Grace is driving an experimental car that can accelerate from 0 m/s to 53.8 m/s in 6.0 s. If the combined mass of Grace and the car is 2000 kg, what is the average power that the engine outputs? Assume there are no drag forces.
ANSWER: _____
20. A team of fourteen huskies pulls Scott and his sled across snow at 4.1 m/s. Scott and his loaded sled have a combined mass of 320 kg and the average mass of each dog is 18.9 kg. If each dog can pull with a force of 190 N, what is the magnitude of the acceleration of the sled? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.
ANSWER: _____
21. Nicholas works in a factory sorting packages on a roller-belt conveyor system. Feeling a bit impatient, he helps move a 13.3-kg package along by pushing it over a distance of 3.2 m with a constant force of 80 N. Before he touched the package, it was moving at a constant speed of 0.4 m/s. If the opposing friction force between the conveyor belt and the package averages 6.4 N, calculate the final velocity of the package.
ANSWER: _____
22. Patricia is an astronaut working outside the International Space Station (ISS) where the atmospheric pressure is essentially zero. The pressure gauge on the astronaut's air tank reads $5.2 \times 10^6 \text{ Pa}$. What force does the air inside the tank exert on the flat end of the cylindrical tank, a disk that is 0.22 m in diameter?
ANSWER: _____

23. Ian inflates his bicycle tire to an absolute pressure of 0.9 MPa (a gauge pressure of about 115 psi) at a temperature of 6.7°C. What is the tire's gauge pressure after the temperature has risen to 29.8°C? Assume there are no leaks and the volume of the tire does not change.
- ANSWER: _____
24. Jonathan places an outdoor sandwich-board advertising sign on the ground. The sign used to have a chain 0.56 m vertically below its hinge, but the chain is broken. The sign's mass is 8 kg. The sign has a vertical height of 1.76 m. The hinge is opened so that the feet of the sign measure 2.9 m apart. What is the minimum coefficient of static friction between the feet of the sign and the sidewalk? To draw the sign, you may draw a large A. The chain, now broken, would have resembled the horizontal line in the A and the hinge is at the top of the A.
- ANSWER: _____
25. Thomas is trying to keep a 40.0-kg wagon at rest on a frictionless inclined plane. The plane makes an angle of 76.1° with the horizontal. What is the magnitude of the force Thomas must apply on the wagon to keep it from sliding down the plane?
- ANSWER: _____
26. Zoe drives a 2190-kg tractor that pulls a 490-kg log along the ground for 40 m. The rope connecting the tractor to the log makes an angle of 21° with the ground and is acted on by a tensile force of 4000 N. How much work does Zoe's tractor perform during this motion?
- ANSWER: _____
27. Stephen is a contestant in a winter sporting event. He is tasked with pulling a rope connected to a 550-kg block of ice on a frozen lake. Stephen pulls diagonally upward on the rope at an angle of 55° from the vertical. What is the minimum force at which he must pull in order to get the block of ice moving? Note that for ice on ice, $\mu_s = 0.1$ and $\mu_k = 0.03$.
- ANSWER: _____
28. Adam, a fireman by profession but physicist at heart, jumps horizontally from a burning building with an initial speed of 9.1 m/s. Later, Adam is interested in determining the time at which the angle between his velocity and acceleration vectors is 43°. At what time does that happen?
- ANSWER: _____
29. David is building a seesaw for his daughter, Alexis, and plans on making it 14 ft long. He plans to sit on one side, 5.9 ft away from the center of the seesaw, and his daughter will sit on the other side, 5.9 ft away from the center of the seesaw. David has weighed himself, his daughter, and the seesaw and learned that his weight is 148 lbs, his daughter's weight is 70 lbs, and the weight of the seesaw is 13.2 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Alexis can cause a rotation in the seesaw just as easily as himself. Where must the fulcrum be placed?
- ANSWER: _____
30. Kyle is designing a new 2.8-m-long, 20-kg diving board for DiveMaster, Inc. He bolted a prototype into the ground 2.4 m away from the diving edge with its spring being 2.0 m away from the diving edge. Kyle notes that the spring compresses by 2 cm at equilibrium under the weight of the diving board alone. When Kyle, who has a mass of 83 kg, gets on the prototype diving board at a distance of 0.8 m from the diving edge, by how much does the spring compress at equilibrium? Report your answer in centimeters.
- ANSWER: _____
31. Matthew, a 83-kg ice-hockey goalie who is originally at rest, catches a 0.18-kg ice-hockey puck that is traveling at a velocity of 42.7 m/s. What is the net change in Matthew's and the puck's kinetic energy during the collision? Assume friction is negligible.
- ANSWER: _____
32. Elizabeth is a design engineer for a lightbulb company. She manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 3°C. What is the pressure in the light bulb when it rises to 234°C? The coefficient of volume expansion for the glass light bulb is $34 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$.
- ANSWER: _____
33. Patricia puts a coin on a scale and finds its mass is 8.03 g. When the coin and scale are completely submerged in water, the scale reports the coins mass as 1.03 g. What is the density of the coin?
- ANSWER: _____
34. Find the increase in entropy as 1.6 kg of ice, originally at 0.9°C, melts to form water at 0.9°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.
- ANSWER: _____

35. Two identical billiard balls strike a rigid wall with the same speed and bounce without any change of speed. The first ball strikes perpendicular to the wall. The second ball strikes the wall at an angle of 11.2° from the perpendicular, and bounces off at an angle of 11.2° from perpendicular to the wall. Calculate the ratio of the magnitudes of impulses on the two balls from their collision with the wall.

ANSWER: _____

36. If a marathon runner averages 10 mi/h, how long does it take her to run a 25.7-mi marathon?

ANSWER: _____

37. Calculate the work output of a Carnot engine operating between temperatures of 539 K and 362 K for 6090 J of heat transfer to the engine.

ANSWER: _____

38. As Ian waits for his food to heat up, he determines that his microwave oven rotates the food with a speed of 14.2 revolutions/minute. What is the food's angular velocity in radians per second?

ANSWER: _____

39. Ethan's 65-L (17.2-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 2.2°C . How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 39.8°C ? The coefficients of volume expansion for steel and gasoline are respectively $35 \times 10^{-6} \text{ } 1/^\circ\text{C}$ and $950 \times 10^{-6} \text{ } 1/^\circ\text{C}$.

ANSWER: _____

40. Jason is roller skating, skating over a 0.5-m-tall hill. Assuming that he is moving at a speed of 5.5 m/s at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: _____

41. Nicholas is hanging from a pull-up bar inside an elevator that is accelerating downward at a rate of 2.8 m/s^2 . Nicholas's mass is 85.0 kg and the distance between the elevator floor and the bottom of his feet is 0.64 m. Nicholas lets go of the bar just as the elevator changes its acceleration to 11.2 m/s^2 upward. At what speed will he hit the elevator floor?

ANSWER: _____

42. Joseph launches a 2.0-kg rocket is traveling at a velocity of 6.1 m/s. What is its kinetic energy?

ANSWER: _____

43. A 1.4-kg rock is shot up vertically at the same time that a 0.4-kg ball is projected horizontally. Neglecting friction, if both start from the same height, which hits the ground first? Do they hit the ground at the same time? Describe the motion in words.

ANSWER: _____

44. Nick throws a plastic cup horizontally from his dorm window 17.7 m above the ground. Brian, another student, stands 7.51 m away from the dorm and catches the plastic cup at a height of 1.6 m above the ground. Brian claims that Nick threw the plastic cup too slowly, so Nick wants to calculate its initial speed to prove him wrong. Ignoring friction, what initial speed does he calculate?

ANSWER: _____

45. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL, at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 6.3 mmHg above atmospheric pressure? Recall that $1.0 \text{ mmHg} = 133 \text{ Pa}$.

ANSWER: _____

46. A man walks 178.7 m east and then 68.7 m south. What is the difference between his travelled distance and the absolute value of his displacement?

ANSWER: _____

47. A BASE jumper runs off a cliff with a (horizontal) speed of 1.6 m/s. Which of the following is closest to his overall speed after 1.78 seconds?

ANSWER: _____

48. A particular dam is 800-m wide and contains water that is 40-m deep. What is the average pressure on the dam due to the water?

ANSWER: _____

49. Grace creates a soap bubble and measures its diameter to be 167 μm . Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m. Report your answer in mmHg, recalling that 133 Pa is 1 mmHg.

ANSWER: _____

50. Laika the moon dog jumps 3.94 m into the air. What is the vertical speed at which Laika leaves the ground? Laika is on the Moon and so her free-fall acceleration is 1.625 m/s^2 . Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.

ANSWER: _____

51. A frog just barely jumps over a 1.48-m-tall fence. How long is the frog in the air, measured from when it jumps to when it lands?

ANSWER: _____

52. Alexis is with her daughter, Grace, at a park. Grace has climbed onto a merry-go-round. Alexis is continuously pushing on the 29-kg merry-go-round with a force of 105 N at its edge, 1.4 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 31-kg Grace is 0.56 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: _____

53. On a family trip, Grace and Sam take their son, Ryan, to visit the deepest man-made hole on Earth, the Kola Superdeep Borehole. Ryan shows little interest, however, and spends all his time on his cell phone. Grace, fairly horrified at her son's lack of scientific interest, grabs the phone and throws it down the hole, which is 9 inches in diameter and 12,262 meters deep. Grace manages to hear over Ryan's cries that it hit the bottom in 49.8 seconds. What was the velocity at which Grace threw the cell phone?

ANSWER: _____

54. Suppose that there is heat transfer of 52 J to a system, while the system does 46 J of work. Later, there is heat transfer of 48 J out of the system while 53 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: _____

55. To test the idea of surviving a free-falling elevator, Gabriel conducts an experiment using a string-attached plank and a programmable toy kangaroo. His makeshift elevator falls 24.1 m and the kangaroo jumps 0.0004 s before impact. Previous tests show that these toys can endure impacts below 6.9 m/s. What is the minimum initial speed needed for the kangaroo's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)

ANSWER: _____

56. Chloe drives a 7700-kg truck and descends a 48-m-tall hill at a constant speed. What is the final temperature of 160 kg of brake material if its temperature is 13°C at the top of the hill? Assume the specific heat of the brake material is $700 \text{ J/kg}^\circ\text{C}$. Assume the brakes do all the work to maintain the speed.

ANSWER: _____

57. Nathan, a construction worker, holds up a plank of wood 2.9-m long. His left hand pushes an end of the plank down with a force F_L , while his right hand pushes the plank up with a force F_R at a distance of 0.9 m from the left hand. The plank is held in equilibrium. The plank has a mass of 20 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_L .

ANSWER: _____

58. Allison nudges a 4.6-kg block of wood so that it just begins moving down a long, wooden ramp angled at 27° above the horizontal. The block of wood must travel 29.2 m to reach the bottom. How long does it take for the block of wood to reach the bottom? Do not neglect friction. Note that for wood on wood, $\mu_s = 0.5$ and $\mu_k = 0.3$.

ANSWER: _____

59. A 0.32-kg blue box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary orange box, which has a mass of 0.33 kg. The blue box emerges from the room at an angle of 74° north of east. The speed of the blue box is originally 5.2 m/s and is 2.1 m/s after the collision. Calculate the speed of the orange box after the collision.

ANSWER: _____

60. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Leah's. To get to Leah's, you have to walk 3.94 km west and then 4.13 km north. What is the compass direction of a line connecting your house to hers?

ANSWER: _____

61. A reservoir has a surface area of 40 km^2 and an average depth of 34 m. What mass of water is held behind the dam?

ANSWER: _____

62. It takes 388 seconds for a dog to move 104 m east, then 444 m north, then 369 m west, and finally 142 m south. What is the dog's net displacement? Include both magnitude and direction.

ANSWER: _____

63. Nicholas kicks a soccer ball, initially at rest on the ground, with an initial velocity of 24.0 m/s at a launch angle of 58.2° above the horizontal. The ball flies through the air and lands back on the ground. Calculate its total flight time, assuming that air resistance is negligible.

ANSWER: _____

64. Joseph is designing a set to be shown in a play. A 80-kg actor stands on a 43-kg platform that is suspended by two ropes, one on its left side, and another on its right side. The left rope is attached to a weight of mass m_L through a frictionless pulley; the right rope is attached to a weight of mass m_R through another frictionless pulley. When the weights are released, the platform must descend with an acceleration of 0.1 m/s^2 without rotating. Joseph needs to make the platform 2.3-m long and the actor must stand 1.2-m from the left side. What must m_R be?

ANSWER: _____

65. A typical small rescue helicopter has four blades, each of which is 3.8-m long and has a mass of 45 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1210 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 262 rpm.

ANSWER: _____

66. Rachel goes outside on a summer morning with her tea and sees a perfectly vertical strand of spider web. She watches for a while and eventually a spider climbs up the strand. At this point, invested in her experiment, Rachel takes the spider and puts it on a scale, measuring its mass to be 468 mg, then returns it to its strand of web. What is the tension in the web when the spider is sitting on it?

ANSWER: _____

67. John attaches a nozzle of diameter 5.8 mm to a garden hose of diameter 2.8 cm. If the hose fills up a 2-L bucket in 12.9 seconds, calculate the pressure in the hose assuming level, frictionless flow.

ANSWER: _____

68. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Aaron's aorta is 11.0 L/min and Aaron's aorta has a radius of 11 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 9.9 μm . The speed of blood in the capillaries is about 0.32 mm/s. Calculate the number of capillaries in Aaron's blood circulatory system.

ANSWER: _____

69. Jennifer places a ladder of mass 18 kg against her house. The feet of the ladder are resting on her concrete patio at a distance of 7.9 m from the house. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 6.4 m above the concrete patio. The ladder's center of gravity is at its midpoint. Jennifer is standing on a rung at a horizontal distance of 2.8 m from the house. If Jennifer's mass is 57 kg, what is the magnitude of the friction force between the feet of the ladder and the concrete?

ANSWER: _____

70. A 851-kg rocket ship, travelling at 58 m/s, is acted upon by an average force of 30 kN applied in the direction of its motion for 35 s. What is the change in the rocket's velocity?

ANSWER: _____

71. Chloe pushes on a 35.4-kg package on a roller-belt conveyor system with a constant force of 110 N. The package moves over a distance of 9.2 m and the opposing friction force averages 25.3 N during this motion. Calculate the net work done on the package.
ANSWER: _____
72. Tyler is riding on a roller coaster, ascending a steep track as he approaches the top of a 46.2-m peak. Assuming Tyler is at rest at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.
ANSWER: _____
73. Jason places a 0.1-kg toy car next to a spring on a track. He moves the car backward to compress the spring 5.4 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Jason greased the axles of the toy car, assume there is no friction. Jason's previous experiments indicate that the spring has a force constant of 100 N/m.
ANSWER: _____
74. Zoe's doctor performs a special test and tells her that a 3.3-N force can rupture her eardrum. If her eardrum has an area of 0.9 cm^2 , at what depth in freshwater would her eardrum rupture, assuming the gauge pressure in the middle ear is zero and she does nothing to alleviate the pressure?
ANSWER: _____
75. Robert is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 399 N through a rotation of 0.57 rad (33°)? He exerts the force perpendicular to the grindstone's 60-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.
ANSWER: _____
76. A grey heron is flying horizontally at a speed of 26.1 m/s when the trout in her talons wiggles loose and falls into the lake 1.2 m below. Calculate the angle at which the fish hits the water.
ANSWER: _____
77. Chloe flips her bicycle upside down and moves the pedals so that the rear wheel begins spinning. The wheel starts from rest and reaches a final angular velocity of 184 rpm in 3.1 s. Chloe then slams on the brakes to cause an angular acceleration of -54 rad/s^2 . How long does it take the wheel to stop?
ANSWER: _____
78. Lauren is a 89-kg athlete competing in a race. She begins from rest with an average acceleration of 1.9 m/s^2 for 1.1 s. What is her momentum at 1.1 s?
ANSWER: _____
79. Emma is not only a part-time guide at the Battleship USS Iowa Museum, but also a physics student. She wants to find the net external force that would be exerted on a 900-kg artillery shell fired from one of the USS Iowa's turrets. Emma measures the barrel of a turret to be 18.8-m long and reads that a shell, fired from rest, would leave the barrel at 590 m/s. What is the net external force exerted on the shell?
ANSWER: _____
80. Grace aims her arrow towards a target located 235.3 m away. The target's bull's eye aligns with the height from which Grace releases the arrow. The arrow is launched at an angle with an initial speed of 79.3 m/s. Find the angle at which Grace launched the arrow.
ANSWER: _____
81. Zoe skateboards up a curved ramp (quarterpipe) and launches herself into the air. She begins her motion with a speed of 11.4 m/s at a total distance of 4.6 m away from the launch position. In order to launch herself from the top of the quarterpipe, Zoe does 340 J of work. Zoe has a mass of 72 kg and knows that the average friction force acting on her as she skates across this particular surface is 22 N. If the quarterpipe has a height of 0.5 m, how far above the quarterpipe does Zoe reach?
ANSWER: _____
82. David throws a 3.0-g coin straight downward from a 26-story window with a vertical speed of 13.5 m/s (approximately 30.2 mph). Ignoring the effect of the drag force, at what speed does the coin impact a patch of grass on the ground? Express the speed in meters per second. Take a story as roughly 10.3 feet. There are 3.28 feet in one meter.
ANSWER: _____

83. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Isabella's aorta if the flow rate of her blood is 9.1 L/min. Isabella's aorta has a radius of 11 mm.
ANSWER: _____
84. Calculate the maximum buoyant force that water could exert on a ship with a mass of 2.0×10^8 kg if the ship displaces 7.1×10^6 m³ of water.
ANSWER: _____
85. Michael is watching his dog, Champ, play around in the backyard. Champ is standing still until he sees a squirrel. Over 5.3 s, Champ covers a distance later measured to be 35.2 m. Michael is curious: how fast was Champ moving at 5.3 s?
ANSWER: _____
86. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1277-m long on the coldest day. The bridge was exposed to temperatures ranging from -6°C to 36°C. What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is 12×10^{-6} 1/°C.
ANSWER: _____
87. Lauren puts a 0.4-kg aluminum pot on the stove and fills it with 0.65 liters of water. Both the pot and the water are initially at 0.2°C. How much heat is required to bring them both to 60.9°C? The specific heat values of water and aluminum are 4186 J/kg°C and 900 J/kg°C respectively.
ANSWER: _____
88. Robert drives his 1530-kg car around a 400-m radius unbanked curve at 33.7 m/s (about 75 mph). Robert's mass is 120 kg. What is the magnitude of the centripetal acceleration of Robert's car? For reference, accelerations over approximately 0.3 m/s² are mildly uncomfortable.
ANSWER: _____
89. A small airplane flies in a straight line at a speed of 208 km/h. How long does it take the plane to fly 916 km?
ANSWER: _____
90. Kevin aims his arrow towards a target located 265.2 m away. The target's bull's eye aligns with the height from which Kevin releases the arrow. Launched at an angle with an initial speed of 70.5 m/s, the arrow passes a thin tree branch positioned directly between Kevin and the target. The branch is 156.5 m away from Kevin and is positioned 14.4 meters above the arrow's launch point. What is the vertical distance at which the arrow passes by the branch? A positive value indicates it passed above the branch, while a negative value indicates it passed below.
ANSWER: _____
91. At the end of his foot race, Robert decelerates from a velocity of 5.8 m/s until he comes to rest, taking 3.2 s to do so. How far does Robert travel, measuring from the end of his race until he comes to rest?
ANSWER: _____
92. Jacob drives his 1470-kg car around a 200-m radius unbanked curve at 36.0 m/s (about 80 mph). Jacob's mass is 102 kg. Calculate the centripetal force exerted on the car as Jacob drives around the curve.
ANSWER: _____
93. Robert's 67-L (17.7-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 5.5°C. After both the tank and the gasoline warm to 37.3°C, what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficients of volume expansion for steel and gasoline are respectively 35×10^{-6} 1/°C and 950×10^{-6} 1/°C. Assume the bulk modulus B for gasoline is 1.00×10^9 N/m².
ANSWER: _____
94. Efe, who has a mass of 96.7 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates upward at a rate of 1.5 m/s², what will the scale read?
ANSWER: _____
95. When rebuilding his car's engine, Nick finds that he must exert 390 N of force to insert a dry steel piston into a steel cylinder. What is the magnitude of the normal force between the piston and cylinder as the piston is inserted into the cylinder at a constant speed? Note that for dry steel on dry steel, $\mu_s = 0.6$ and $\mu_k = 0.3$.
ANSWER: _____

96. Two children, Kevin and Abigail, and a toy boat are balancing on a long seesaw of negligible mass. Kevin has a mass of 15 kg and sits 3.8 m away from the fulcrum. Abigail, who has a mass of 17 kg, sits on the other side of the fulcrum, opposite to Kevin. The toy boat has a mass of 3.7 kg and is on Abigail's side of the seesaw, a distance of 2.2 m away from the fulcrum. What is Abigail's distance from the fulcrum?

ANSWER: _____

97. Calculate the depth below the surface of water at which the pressure due to the weight of the water equals 0.29 atm.

ANSWER: _____

98. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 4880 J of heat transfer occurs from a hot reservoir at 667 K to a cold reservoir at 234 K, assuming that there is no temperature change in either reservoir.

ANSWER: _____

99. Chloe is with her daughter, Elizabeth, at a park. She is continuously pushing on a 34-kg merry-go-round with a force of 56 N at its edge, 1.2 m away from the center. Elizabeth watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: _____

100. Paul is skiing at a speed of 11.6 m/s just as he crests a hill. On the downslope, he accelerates at 3.8 m/s^2 for 7.0 s. Calculate his speed at the end of the 7.0 s.

ANSWER: _____

101. Jennifer flips her bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 302 rpm in 3.0 s, calculate the angular acceleration in rad/s^2 .

ANSWER: _____

102. What is the average kinetic energy of a gas molecule at 164°C ?

ANSWER: _____

103. Suppose that Adam floats in freshwater with 93.9% of his volume submerged when his lungs are full of air. His mass is 86 kg. What is his average density?

ANSWER: _____

104. A rocket with a mass at liftoff of 3.1 million kg burns fuel at a rate of 18000 kg/s. Its exhaust velocity was measured to be 2600 m/s. Calculate its acceleration at liftoff.

ANSWER: _____

105. Joshua begins spinning a large grindstone from rest by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 261 N through a rotation of 1.23 rad (70°). He exerts the force perpendicular to the grindstone's 32-cm radius and the effects of friction are negligible. If the grindstone's mass is 58 kg, what is its final rotational kinetic energy? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: _____

106. Brian, a 81-kg ice-hockey goalie who is originally at rest, is hit by a 0.17-kg ice-hockey puck that is traveling at a velocity of 38.1 m/s. Brian hits the puck in such a way that this may be considered an elastic collision. Calculate the speed of the goalie after he hits the puck.

ANSWER: _____

107. Natalie is sitting in her folding chair on the beach, watching her friend Nick as he rides in a sailboat moving at a velocity of 5.47 m/s at an angle of 16.1° North of East. Nick then passes his brother, Shiv, who is riding in a yacht, which, from Nick's perspective, is travelling at a velocity of 2.32 m/s at an angle of 73.3° South of East. From Natalie's perspective, what is the velocity of Shiv on the yacht?

ANSWER: _____

108. Jennifer is producing heat at a rate of about 160 W while doing light exercise. At what rate must water evaporate from her body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C .

ANSWER: _____

109. Two nurses, Chloe and Zoe, work together to push a crash cart by pushing on the handle. They push it from rest so that it covers a distance of 12.9 m in 4.1 s. Chloe exerts 110 N at 44.0° below the horizontal and Zoe exerts 130 N at 26.9° below the horizontal. The force of friction is 72 N. Determine the mass of the cart.

ANSWER: _____

110. Natalie drops a snowball off a cliff and measures it striking the ground with a vertical impact velocity of 70.9 m/s. Determine the height of the cliff. Report your answer in feet, recalling that 3.28 ft = 1 m.

ANSWER: _____